HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Department of Computer Science and Engineering

BSC(Honors) in CSE

ENG 127: English Skills

Final Examination

Spring 2025

2nd Semester

September 08, 2025 Time: 2 Hours

Total Marks: 40

No.	Question/ Statement	CLOs	Marks
1	**Some people think that governments should take the complete responsibility to keep our environment clean. Others believe that it is the responsibility of individual citizens. Discuss both views and give your own opinion.**	CLO4	12×1=12
2.	**Write a review of a book you have read recently. In your review, mention:** A brief summary of the story/content The main themes/ideas of the book Your personal evaluation and recommendation	CLO4	08×1=08
3.	**Write a formal letter to the manager of a company applying for an internship position.**	CLO4	10×1=10
4.	**Paraphrase the following in your own words:** Computers have changed the way people live and work in the modern world. They are used for writing, designing, calculating, and even entertainment. With the help of the internet, computers connect people across the globe within seconds. They also save time by performing difficult tasks faster than humans. As a result, computers have become a vital part of our daily lives.	CLO2	5×1=5
5.	**Define the difference between a formal letter and an informal letter with examples.**	CLO1	5×1=5

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
EEE 127: Electronic Devices and Circuits

Batch: 27^{th} | Semester: Spring 2025

Time: 2 Hour

Date: 15/09/2025 | **Final Examination** | Total Marks: 40

Answer any **five** questions: $5 \times 8 = 40$

1.	(a)	Why do we use Zener Diode?	CLO3	2
	(b)	What is the difference between Silicon diode and Zener Diode?	CLO2	2
	(c)	Determine the minimum and maximum voltage drop across a resistor for Zener diode.	CLO2	4
2.	(a)	Calculate the change in Zener voltage if Zener voltage is 15V at room temperature and new Temperature is 1000° C. Temperature co-efficient (Tc) is 0.072%.	CLO2	4
	(b)	Determine the range of voltage (V) to maintain the Zener Diode on state.	CLO3	4
3.	(a)	What is Clipper Circuit?	CLO3	2
	(b)	Demonstrate the output signal of a digital pulse (clock signal) after clipping for silicon diode.	CLO3	3
	(c)	Consider a silicon diode with a Sinusoidal wave as input, if the voltage is 0.7V then show the output voltage curve for both positive and negative half cycle.	CLO3	3
4.	(a)	What is Threshold Voltage?	CLO1	2
	(b)	Explain the Field Effect Transistor.	CLO1	3
	(c)	Write the differences between JFETs and BJTs.	CLO1	3
5.	(a)	What is the Pinch Off Voltage?	CLO1	2
	(b)	Explain Full wave Bridge Rectification with the necessary circuit diagram.	CLO2	3
	(c)	Derive the equations for Average voltage and RMS voltage in a Half-Wave Rectifier.	CLO2	3
6.	(a)	Draw the Construction Circuit for both DE-MOSFET and E-MOSFET.	CLO3	4
	(b)	Draw and Explain the output Characteristic Curve for DE-MOSFET and E-MOSFET	CLO3	4

7.	(a)	Write four Applications of FET.	CLO3	3
	(b)	Draw and briefly discuss the Construction Circuit and Output Characteristic curve for n-channel JFET.	CLO3	5

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 121: Structured Programming Language

Batch: 27th | Semester: Spring 2025

Date: 10/09/2025 | **Final Examination** | Time: 2 Hours | Total Marks: 40

Answer any five questions. $5 \times 8 = 40$

			CLO	Marks
1.	(a)	Define Array. Explain how arrays have simplified the declaration process and manipulation of data.	CLO3	4
	(b)	Write a C program that sorts the elements of a one-dimensional array in ascending order.	CLO4	4
2.	(a)	Define loop. Differentiate between i. Counter-controlled and Sentinel-controlled loop ii. Entry-controlled and Exit-controlled loop	CLO3	4
	(b)	Write a C program that takes an integer input from the user and reverses the digits of the number using a while loop. The program should then display the reversed number as output. (e.g. 1234 → 4321)	CLO4	4
3.	(a)	Briefly explain the key componeents of a user-defined function in C programming.	CLO4	3
	(b)	What are recursive functions? Write a C program that calculates the factorial of a given positive integer using recursion. The program should take an integer as input and return its factorial.	CLO3 & CLO4	5
4.	(a)	Explain the differences between call by value and call by reference with an example.	CLO3	2
	(b)	Find out the output of the following programs –	CLO5	3×2=6

i.
```c
#include<stdio.h>
int main()
{
  int a = 8;
  int *p = &a;
  *p = *p + 2;
  printf("%d %d", a, *p);
  return 0;
}
```

ii.
```c
#include<stdio.h>
int main()
{
  int m = 15, *n, *o;
  // Assume address of m is 2004
  n = &m;
  o = n;
  *n -= 3;
  *o += 5;
  m *= 2;
  printf("m = %d, n = %d, o = %d\n", m, n, o);
  return 0;
}
```

iii.
```c
#include<stdio.h>
int main()
{
  int x = 12, y = 18;
  int *ptr1 = &x, *ptr2 = &y;
  *ptr1 += *ptr2;
  *ptr2 = *ptr1 - *ptr2;
  *ptr1 = *ptr1 - *ptr2;
  printf("x = %d, y = %d\n", x, y);
  return 0;
}
```

5. (a)	State the differences between structure and union in C with their syntaxes.	CLO4	3
(b)	Write a C program that defines a structure named Student to store a student's details, including name, roll number, and marks. The program should take input for these details from the user using a loop and then display them.	CLO4	5
6. (a)	Execute the following loop conversions:	CLO5	2×2=4

i. The following for loop prints the multiplication table of 5. Convert it into a ~~for loop~~. while loop

```c
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) {
        printf("5 x %d = %d\n", i, 5*i);
    }
    return 0;
}
```

ii. The following C program calculates the factorial of the number n using a while loop. Convert it into a do-while loop.

```
#include <stdio.h>
int main() {
   int n = 5, factorial = 1, i = 1;
   while (i <= n) {
      factorial *= i;
      i++;
   }
   printf("Factorial = %d\n", factorial);
   return 0;
}
```

(b) Define what a pointer is in C programming language. Explain the reference(&) and dereference(*) operators used with pointers in C. CLO4 4

7. (a) Explain the following functions: CLO4 3

i. strcpy()
ii. strcmp()
iii. strlen()

(b) Write a C program to print the following number pattern based on user input for the number of rows. For example, when row = 4, CLO4 5

```
1
1 2
1 2 3
1 2 3 4
```

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology
Semester Final Examination 2025

Course Code: MAT125 | **Course Title: Linear Algebra and Complex Variable**
Final Examination | Semester: Spring 2025
Date: 9/17/2025 | Total Time: 2:00 Hours | **Total Marks: 40**

Answer any four (04) out of the following questions: $4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No.		Questions	CLOs	Marks
Q1.	(a)	Define complex number.	CLO1	[1]
	(b)	Describe the region determine by the following relations: (i) $\left\|\frac{z-3}{z+3}\right\| = 2$. (ii) $1 < \|z+i\| \leq 2$.	CLO2	[3]
	(c)	State De Moiver's theorem and find all values of the following: (i) $x^7 + x^4 + x^3 + 1 = 0$. (ii) $(x+1)^5 + (x-1)^5 = 0$.	CLO3	[6]
Q2.	(a)	Define analytic function.	CLO1	[1]
	(b)	State and prove the necessary conditions for a function $f(z)$ to be analytic in a region.	CLO2	[5]
	(c)	Show that the function $f(z) = \begin{cases} \frac{x^3-3xy^2+i(y^3-3x^2y)}{x^2+y^2}; & when\ z \neq 0 \\ 0; & when\ z = 0 \end{cases}$, is continuous and Cauchy-Riemann equations are satisfied but $f(z)$ does not differentiable at $z = 0$.	CLO3	[4]
Q3.	(a)	Define harmonic and harmonic conjugate function.	CLO1	[2]
	(b)	Prove that the function $f(z) = e^{z^2}$ satisfied Cauchy-Riemann equations.	CLO2	[4]
	(c)	Show that the function $u = x^3 + 6x^2y - 3xy^2 - 2y^3$ is a harmonic function. Also find harmonic conjugate of u.	CLO3	[4]
Q4.	(a)	Define complex integration.	CLO1	[2]
	(b)	State and prove Cauchy's integral formula.	CLO2	[4]
	(c)	Evaluate $\oint_c \frac{Sin^6 z}{(z-\frac{\pi}{4})^4} dz$, where c is the circle $\|z\| = 1$.	CLO3	[4]
Q5.	(a)	Define singular point, isolated singularities and branch point.	CLO1	[2]
	(b)	Locate and name all the singularities of $f(z) = \frac{z^{10}+z^5+5}{(z-2)^5(3z+5)^2}$.	CLO3	[4]
	(c)	Calculate $\int_2^{4-2i} (z^3 - 4iz + 6) dz$.	CLO3	[4]
Q6.	(a)	State Taylor and Laurrent series.	CLO1	[2]
	(b)	Find the limit and continuity of $f(z) = \begin{cases} z^2; & \text{when } z \neq i \\ 0; & \text{when } z = i \end{cases}$ at $z = -i$.	CLO3	[4]
	(c)	Expand $f(z) = \text{Sinz}$ using Taylor series expansion at $z = \frac{\pi}{4}$ and determine the region of convergence of this series.	CLO3	[4]

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Final Examination (Spring'25)

PHY 111 Electricity & Magnetism / PHY 121 Physics II/ PHY 121 Physics II

(Mechanics, Electricity, Magnetism and Modern Physics)

Time: 2 hours | Marks: 40

Answer any 4 (Four) of the following sets of questions

1. (a) Write short note on the following terms. 2 CLO1

(i) Magnetic Field.

(ii) Toroid.

(b) A charged particle is projected into a magnetic field in a direction perpendicular to the field. Show that the particle moves in a circular path. Obtain expressions for the radius of the circular path and the period for one revolution. Does the period depend on the velocity of the particle? 4 CLO2

(c) An electron is moving with a speed of 5×10^{7} m/s at right angle to a magnetic field of 5500 G. 4 CLO3

(a) What is the magnetic force on the electron?

(b) What is the radius of the circle in which the electron moves?

2. (a) What is Hall effect? 2 CLO1

(b) Show how Hall effect can be used to determine the nature of the charge carriers and the number of charge carriers per unit volume of a conductor. 4 CLO2

(c) A solenoid is 1.0 meter long and 3.0 cm in mean diameter. It has five layers of windings of 850 turns each and carries current of 5.0 amp. 4 CLO3

(a) What is B at its center?

(b) What is the magnetic flux Φ_B for a cross-section of the solenoid at its center?

3. (a) Define the following terms 2 CLO1

(i) Current Density

(ii) Drift Velocity

(b) Show that the relation between current density and drift velocity is, 4 CLO2

$J = -nev_d$.

(c) A nickel wire of length one meter and diameter 0.55 cm has resistance 2.87×10^{-3} ohms. If an e.m.f. of one volt is applied between two ends of the wire, calculate (i) current through the wire and (ii) current density. 4 CLO3

4. (a) State the basic postulates of Einstein's special theory of relativity. 2 CLO1

(b) Derive the Lorentz space time transformation formulae. Show that for values of v << c, Lorentz transformation reduces to the Galilean one. 4 CLO2

	(c)	The proper length of a rod is 5 meters. What would be its length for an observer if it be moving relative to him in a direction parallel to its own length (i) with velocity 0.8c, (ii) moving with velocity c?	4	CLO3
5.	(a)	What is the photo-electric effect?	2	CLO1
	(b)	Establish Einstein's photo-electric equation and show that the maximum velocity of the emitted electrons depends upon the frequency of the incident radiation and not on it's intensity.	4	CLO2
	(c)	The photo-electric threshold of copper is 3200 Å. If ultraviolet light of wavelength 2500 Å falls on it, find (a) the maximum kinetic energy of the photoelectrons ejected, (b) the maximum velocity of the photoelectrons.	4	
6.	(a)	Define the following terms (i) De-Broglie matter waves (ii) Uncertainty principle	2	CLO1
	(b)	Obtain an expression for the de Broglie wavelength of matter waves.	4	CLO2
	(c)	An electron initially at rest is accelerated through a p.d. of 5000V. Compute (i) the de Broglie wavelength and (ii) the wave number of the electron wave.	4	CLO3